Chapter 8. Universal Approximation, Barron's Theorem and Generalization

Lectures 10/21, 10/26: measures, normed spaces, Cybenko's theorem; Barron's theorem and proof; Rademacher complexity

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1 Three Sources of Error

Target f^*, architecture class $\mathcal{F}$, best element $f_{\text{app}} \in \mathcal{F}$, empirical risk minimizer f_{gen}, optimizer output f_{opt}:

$$\underbrace{f^* \to f_{\text{app}}}_{\text{approximation}} + \underbrace{f_{\text{app}} \to f_{\text{gen}}}_{\text{generalization (estimation)}} + \underbrace{f_{\text{gen}} \to f_{\text{opt}}}_{\text{optimization}}.$$

Here: approximation error of one-hidden-layer networks (universality; Barron rate).

2 Preliminaries from Analysis

2.1 Measures

Definition 8.1: measures on a compact set $K \subset \mathbb{R}^n$

The **Borel sets** of K form the smallest collection containing all open subsets of K and closed under complements in K and countable unions. A (nonnegative) **measure** μ assigns $\mu(A) \in [0, \infty]$ to Borel sets with

$$\mu(\emptyset) = 0, \qquad \mu\Big(\bigcup_{j=1}^{\infty} A_j\Big) = \sum_{j=1}^{\infty} \mu(A_j) \quad \text{for pairwise disjoint } A_j.$$

A **signed measure** is the same with real (possibly negative) values, e.g. a difference of two finite measures. $\mu = 0$ means $\mu(A) = 0$ for every Borel set A. $M(I_n)$ denotes the finite signed regular Borel measures on the unit cube $I_n = [0, 1]^n$.

2.2 Normed spaces and density

Definition 8.2: linear and normed spaces

A real **linear space** is closed under addition and real scaling with the vector-space rules, e.g. $C(K) = \{f : K \to \mathbb{R} \text{ continuous}\}$. A **normed linear space** is a linear space X with $\|\cdot\| : X \to [0, \infty)$ such that

$$\|u\| = 0 \iff u = 0, \qquad \|cu\| = |c|\|u\|, \qquad \|u + v\| \le \|u\| + \|v\|.$$

On $C(K)$ we use the sup norm $\|f\|_\infty = \max_{x \in K} |f(x)|$. A subset $D \subseteq X$ is **dense** in X if for every $f \in X$ and $\varepsilon > 0$ there is $g \in D$ with $\|f - g\| < \varepsilon$.

Density in $(C(I_n), \|\cdot\|_\infty)$ is uniform approximation.

3 Cybenko's Universal Approximation Theorem

Motivation. One sigmoidal hidden layer approximates any continuous function uniformly (an existence proof via Hahn–Banach and Riesz).

Definition 8.3: sigmoidal and discriminatory

$\sigma : \mathbb{R} \to [0,1]$ is **sigmoidal** if $\lim_{t\to-\infty}\sigma(t) = 0$ and $\lim_{t\to+\infty}\sigma(t) = 1$. For $\mu \in M(I_n)$, σ is **discriminatory** if

$$\int_{I_n} \sigma(\mathbf{w}^\top \mathbf{x} + \theta)\, d\mu(\mathbf{x}) = 0 \quad \text{for all } \mathbf{w} \in \mathbb{R}^n,\ \theta \in \mathbb{R} \qquad \Longrightarrow \qquad \mu = 0.$$

Theorem 8.4: Cybenko 1989 (UAT)

Let σ be any continuous sigmoidal function. Then finite sums

$$G(\mathbf{x}) = \sum_{j=1}^{N} \alpha_j\, \sigma(\mathbf{w}_j^\top \mathbf{x} + \theta_j), \qquad \mathbf{w}_j \in \mathbb{R}^n,\ \alpha_j, \theta_j \in \mathbb{R},\ N \in \mathbb{N},$$

are dense in $C(I_n)$: for every continuous f and $\varepsilon > 0$ some G satisfies $\sup_{\mathbf{x}\in I_n} |f(\mathbf{x}) - G(\mathbf{x})| < \varepsilon$.

Ingredients: Theorems 8.5 (Hahn–Banach), 8.6 (Riesz), 8.9 (discriminatory).

3.1 The functional-analytic lemmas

Theorem 8.5: a consequence of Hahn–Banach

Let W be a linear, non-dense subspace of a normed linear space X. Then there is a bounded linear functional L on X with $L \neq 0$ on X and $L|_W = 0$.

Proof. Pick $x_0 \notin \overline{W} \neq X$; $d = \inf_{w\in\overline{W}} \|x_0 - w\| > 0$. On $\overline{W} \oplus \mathbb{R}x_0$ let $\ell(w + tx_0) = t$: linear, $\ell|_W = 0$, $\ell(x_0) = 1$, and $|t|d \le |t|\|x_0 + w/t\| = \|w + tx_0\|$ $(t \neq 0)$, so $\|\ell\| \le 1/d$. Extend by Hahn–Banach. ∎

Theorem 8.6: Riesz representation theorem

Let $K \subset \mathbb{R}^n$ be compact. For every bounded linear functional F on $C(K)$ there is a unique finite signed (regular Borel) measure μ on K with

$$F(f) = \int_K f(\mathbf{x})\, d\mu(\mathbf{x}) \qquad \text{for all } f \in C(K).$$

(For positive functionals the measure is nonnegative.)

Theorem 8.7: an annihilating measure

Let W be a linear, non-dense subspace of $C(I_n)$. Then there is $\mu \in M(I_n)$, $\mu \neq 0$, with $\int_{I_n} h\, d\mu = 0$ for all $h \in W$.

Proof [Supplement]. Theorem 8.5 gives bounded $L \neq 0$, $L|_W = 0$; Riesz gives $L(f) = \int f\, d\mu$ with $\mu \neq 0$, and $\int h\, d\mu = L(h) = 0$ on W. ∎

3.2 Density for discriminatory functions

Theorem 8.8: density for discriminatory activations

If σ is continuous and discriminatory, the sums G above are dense in $C(I_n)$.

Proof. $W = \{G\}$ is a linear subspace of $C(I_n)$. If not dense, Theorem 8.7 gives $\mu \neq 0$ annihilating W, in particular

$$\int_{I_n} \sigma(\mathbf{w}^\top \mathbf{x} + \theta)\, d\mu(\mathbf{x}) = 0 \qquad \text{for all } \mathbf{w} \in \mathbb{R}^n,\ \theta \in \mathbb{R}.$$

so $\mu = 0$ (discriminatory): contradiction. ■

3.3 Continuous sigmoidals are discriminatory

Theorem 8.9: continuous sigmoidals are discriminatory

Every continuous sigmoidal function is discriminatory for all $\mu \in M(I_n)$.

Proof [Supplement]. Assume $\int \sigma(\mathbf{w}^\top \mathbf{x} + \theta)\, d\mu = 0$ for all $\mathbf{w}, \theta$. Fix $\mathbf{w}, \theta, \varphi$ and let $\sigma_\lambda(\mathbf{x}) = \sigma(\lambda(\mathbf{w}^\top \mathbf{x} + \theta) + \varphi)$. As $\lambda \to \infty$, $\sigma_\lambda \to 1$ on $\Pi = \{\mathbf{w}^\top \mathbf{x} + \theta > 0\}$, $\to 0$ on $\{\mathbf{w}^\top \mathbf{x} + \theta < 0\}$, $= \sigma(\varphi)$ on $H = \{\mathbf{w}^\top \mathbf{x} + \theta = 0\}$; by dominated convergence ($|\sigma_\lambda| \leq 1$)

$$0 = \int \sigma_\lambda \, d\mu \ \longrightarrow\ \mu(\Pi) + \sigma(\varphi)\mu(H) \qquad \text{for every } \varphi.$$

$\varphi \to \mp\infty$ gives $\mu(\Pi) = \mu(H) = 0$ for all half-spaces and hyperplanes. Fix $\mathbf{w}$; $F(h) = \int h(\mathbf{w}^\top \mathbf{x})\, d\mu(\mathbf{x})$ is bounded linear on bounded measurable h, with $F(\mathbf{1}_{[\theta,\infty)}) = \mu(\Pi) + \mu(H) = 0$ and $F(\mathbf{1}_{(\theta,\infty)}) = 0$; so F vanishes on interval indicators, simple functions, and by uniform approximation all bounded measurable h. With $h = \cos, \sin$: $\int e^{i\mathbf{w}^\top \mathbf{x}}\, d\mu = 0$ for all $\mathbf{w}$, so $\mu = 0$. ■

Proof of Cybenko's theorem. Theorem 8.9 shows σ is discriminatory; Theorem 8.8 gives density. ■

Remark 8.10: what the theorem says and does not say

Any compact K works after rescaling. Pure existence: no bound on N, no algorithm, no generalization. Extends to every continuous non-polynomial activation incl. ReLU (Leshno et al. 1993). 1D intuition: steep sigmoid $\approx$ step, two steps make a bump, bumps sum to any continuous function.

4 Barron's Theorem: a Dimension-Free Rate

4.1 Fourier preliminaries

For $f \in L^1(\mathbb{R}^d)$ the Fourier transform is $\hat{f}(\boldsymbol{\xi}) = \int_{\mathbb{R}^d} f(\mathbf{x}) e^{-2\pi i \boldsymbol{\xi}^\top \mathbf{x}}\, d\mathbf{x}$. The Sobolev space $H^s(\mathbb{R}^d)$ consists of f with $\int |\hat{f}(\boldsymbol{\xi})|^2 (1 + |\boldsymbol{\xi}|^2)^s\, d\boldsymbol{\xi} < \infty$.

Definition 8.11: Barron class

For $C > 0$,

$$\Gamma_C = \Big\{ f \in L^1(\mathbb{R}^d) : \ \|\hat{f}\|_{L^1(\mathbb{R}^d)} < \infty, \ \int_{\mathbb{R}^d} |2\pi\boldsymbol{\xi}| \, |\hat{f}(\boldsymbol{\xi})| \, d\boldsymbol{\xi} < C \Big\}.$$

On a domain Ω one also uses the spectral Barron norm $\|f\|_{\mathcal{B}^s(\Omega)} = \inf_{f_e|_\Omega = f} \int_{\mathbb{R}^d} (1 + |\boldsymbol{\xi}|)^s |\hat{f}_e(\boldsymbol{\xi})| \, d\boldsymbol{\xi}$, the infimum over all extensions f_e of f beyond Ω.

Theorem 8.12 (Theorem 1, Barron 1993)

Let σ be sigmoidal, $f \in \Gamma_C$ and $B_1^d = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\| \le 1\}$. Then for every $c > 4C^2$ and every $N \in \mathbb{N}$ there is a network f_N with N neurons such that

$$\frac{1}{|B_1^d|} \int_{B_1^d} |f(\mathbf{x}) - f_N(\mathbf{x})|^2 \, d\mathbf{x} < \frac{c}{N}.$$

The rate $O(1/N)$ is free of d; the price is the spectral condition $f \in \Gamma_C$.

Remark 8.13: curse of dimensionality

A fixed linear family needs $\sim \varepsilon^{-d/s}$ terms for s-smooth functions; Barron needs $\approx c/\varepsilon$ neurons for every d (d enters only via C).

5 Proof of Barron's Theorem

Lemma 1 (Maurey): closed-convex-hull elements are approximated at rate $1/N$ by N-term averages. Lemma 2: Barron functions lie in the hull of half-space indicators. Finally indicators become sigmoids.

5.1 Convex hulls and Maurey's lemma

Definition 8.14: convex hull

For a subset G of a vector space,

$$\mathrm{co}(G) = \Big\{ \sum_{j=1}^{n} \alpha_j x_j : \ n \in \mathbb{N}, \ x_j \in G, \ \alpha_j \ge 0, \ \sum_{j=1}^{n} \alpha_j = 1 \Big\},$$

and $\overline{\mathrm{co}}(G)$ denotes its closure in a given norm.

Theorem 8.15 (Lemma 1, Maurey)

Let H be a Hilbert space and $G \subset H$ with $\|g\|_H \le B$ for all $g \in G$, for some $B > 0$. Let $f \in \overline{\mathrm{co}}(G)$. Then for every $N \in \mathbb{N}$ and every $c > B^2$ there exist $g_1, \dots, g_N \in G$ with

$$\Big\| f - \frac{1}{N} \sum_{i=1}^{N} g_i \Big\|_H^2 \le \frac{c}{N}.$$

Proof. Fix N, $\varepsilon > 0$, and $\alpha_j \in [0,1]$ summing to 1, $h_j \in G$ with

$$f^* = \sum_{j=1}^{m} \alpha_j h_j \quad \text{satisfies} \quad \|f - f^*\|_H < \varepsilon.$$

Claim. There are $g_1, \dots, g_N \in \{h_1, \dots, h_m\}$ with $\left\|f^* - \frac{1}{N}\sum_i g_i\right\|_H^2 \le B^2/N$.
Let $X_1, \dots, X_N$ be iid with $P(X_i = h_j) = \alpha_j$, so $\mathbb{E}X_i = f^*$. Expanding,

$$\mathbb{E}\Big\|f^* - \frac{1}{N}\sum_{j=1}^{N} X_j\Big\|^2 = \frac{1}{N^2}\,\mathbb{E}\Big[\sum_{j=1}^{N}\|f^* - X_j\|^2 + \sum_{i\neq j}\langle f^* - X_i, f^* - X_j\rangle\Big].$$

For $i \neq j$, $\mathbb{E}\langle X_i, X_j\rangle = \langle f^*, f^*\rangle$ by independence, so

$$\mathbb{E}\langle f^* - X_i, f^* - X_j\rangle = \langle f^*, f^*\rangle - \langle f^*, \mathbb{E}X_j\rangle - \langle \mathbb{E}X_i, f^*\rangle + \mathbb{E}\langle X_i, X_j\rangle = 0.$$

The diagonal terms are identically distributed:

$$\begin{aligned}\mathbb{E}\Big\|f^* - \frac{1}{N}\sum_j X_j\Big\|^2 &= \frac{1}{N}\mathbb{E}\|f^* - X_1\|^2 = \frac{1}{N}\mathbb{E}[\|f^*\|^2 - 2\langle f^*, X_1\rangle + \|X_1\|^2] \\ &= \frac{1}{N}(\mathbb{E}\|X_1\|^2 - \|f^*\|^2) \le \frac{B^2}{N},\end{aligned}$$

by $\mathbb{E}\langle f^*, X_1\rangle = \|f^*\|^2$, $\|X_1\| \le B$. Some realization $g_i = X_i(\omega)$ is at most the mean: the claim.
By the triangle inequality,

$$\Big\|f - \frac{1}{N}\sum_i g_i\Big\|_H \le \|f - f^*\|_H + \Big\|f^* - \frac{1}{N}\sum_i g_i\Big\|_H < \varepsilon + \frac{B}{\sqrt{N}}.$$

$\varepsilon < (\sqrt{c} - B)/\sqrt{N}$ makes this $< \sqrt{c/N}$ (as $c > B^2$); square. ∎

It remains to put every $f \in \Gamma_C$ in the closed convex hull of affine Heaviside functions.

5.2 Barron functions lie in a convex hull of half-space indicators

Normalized norm on the unit ball:

$$\|g\|_{L_0^2(B_1^d)} = \Big(\frac{1}{|B_1^d|}\int_{B_1^d} |g(\mathbf{x})|^2\, d\mathbf{x}\Big)^{1/2},$$

and scaled half-space indicators

$$G_C = \Big\{\mathbf{x} \in B_1^d \mapsto \gamma\, \mathbf{1}_{\mathbb{R}_+}(\langle \mathbf{a}, \mathbf{x}\rangle + b) : \ \mathbf{a} \in \mathbb{R}^d,\ b \in \mathbb{R},\ |\gamma| \le 2C\Big\},$$

$\mathbf{1}_{\mathbb{R}_+}(t) = \mathbf{1}\{t \ge 0\}$ (Heaviside). Every $g \in G_C$ has $\|g\|_{L_0^2(B_1^d)} \le |\gamma| \le 2C$.

Theorem 8.16 (Lemma 2, Barron 1993)

Let $d \in \mathbb{N}$, $C > 0$ and $f \in \Gamma_C$. Then

$$f|_{B_1^d} - f(\mathbf{0}) \in \overline{\mathrm{co}}(G_C),$$

where the closure is taken in $L_0^2(B_1^d)$.

Proof. $K = \overline{\mathrm{co}}(G_C)$ is closed convex in the Hilbert space $L^2_0(B_1^d)$; $h = f|_{B_1^d} - f(\mathbf{0})$. (The lecture sketched Step 1 and a Riemann-sum idea; parts marked [Supplement] fill the gaps after Barron.)

Step 1 (integral representation). $f, \hat{f} \in L^1(\mathbb{R}^d)$, so by Fourier inversion

$$f(\mathbf{x}) - f(\mathbf{0}) = \int_{\mathbb{R}^d} \hat{f}(\boldsymbol{\xi})(e^{2\pi i\langle \mathbf{x},\boldsymbol{\xi}\rangle} - 1)\, d\boldsymbol{\xi}.$$

With phase $\hat{f}(\boldsymbol{\xi}) = |\hat{f}(\boldsymbol{\xi})|e^{i\kappa(\boldsymbol{\xi})}$,

$$f(\mathbf{x}) - f(\mathbf{0}) = \int_{\mathbb{R}^d} |\hat{f}(\boldsymbol{\xi})|(e^{i(2\pi\langle \mathbf{x},\boldsymbol{\xi}\rangle + \kappa(\boldsymbol{\xi}))} - e^{i\kappa(\boldsymbol{\xi})})\, d\boldsymbol{\xi}.$$

and taking real parts (f real),

$$f(\mathbf{x}) - f(\mathbf{0}) = \int_{\mathbb{R}^d} \frac{\cos\left(2\pi\langle \mathbf{x},\boldsymbol{\xi}\rangle + \kappa(\boldsymbol{\xi})\right) - \cos\kappa(\boldsymbol{\xi})}{|2\pi\boldsymbol{\xi}|}\, |2\pi\boldsymbol{\xi}|\, |\hat{f}(\boldsymbol{\xi})|\, d\boldsymbol{\xi}.$$

(the integrand vanishes at $\boldsymbol{\xi} = \mathbf{0}$). Let $C_f = \int |2\pi\boldsymbol{\xi}||\hat{f}(\boldsymbol{\xi})|\, d\boldsymbol{\xi} < C$ and $d\Lambda(\boldsymbol{\xi}) = C_f^{-1}|2\pi\boldsymbol{\xi}||\hat{f}(\boldsymbol{\xi})|\, d\boldsymbol{\xi}$, a probability measure (lecture: C for C_f; if $C_f = 0$, $h = 0 \in K$). With

$$q_{\boldsymbol{\xi}}(\mathbf{x}) = \frac{\cos\left(2\pi\langle \mathbf{x},\boldsymbol{\xi}\rangle + \kappa(\boldsymbol{\xi})\right) - \cos\kappa(\boldsymbol{\xi})}{|2\pi\boldsymbol{\xi}|}, \qquad k_{\boldsymbol{\xi}} = C_f q_{\boldsymbol{\xi}},$$

we obtain

$$f(\mathbf{x}) - f(\mathbf{0}) = C_f \int_{\mathbb{R}^d} q_{\boldsymbol{\xi}}(\mathbf{x})\, d\Lambda(\boldsymbol{\xi}) = \int_{\mathbb{R}^d} k_{\boldsymbol{\xi}}(\mathbf{x})\, d\Lambda(\boldsymbol{\xi}). \tag{5}$$

Step 2 (each $k_{\boldsymbol{\xi}}$ lies in K). [Supplement]

$k_{\mathbf{0}} = 0 \in G_C$ ($\gamma = 0$). For $\boldsymbol{\xi} \neq \mathbf{0}$, $\mathbf{u} = \boldsymbol{\xi}/|\boldsymbol{\xi}|$, $t = \langle \mathbf{u}, \mathbf{x}\rangle \in [-1, 1]$, $k_{\boldsymbol{\xi}}(\mathbf{x}) = r(t)$ with

$$r(t) = C_f \frac{\cos\left(2\pi|\boldsymbol{\xi}|t + \kappa\right) - \cos\kappa}{2\pi|\boldsymbol{\xi}|}, \qquad r(0) = 0, \qquad |r'(t)| = C_f|\sin\left(2\pi|\boldsymbol{\xi}|t + \kappa\right)| \le C_f,$$

so $|r(s) - r(t)| \le C_f|s - t|$.

Staircase. $t_k = k/m$ ($k = 0, \ldots, m$),

$$s_m(t) = \sum_{k=1}^{m} \alpha_k \mathbf{1}_{\mathbb{R}_+}(t - t_k) + \sum_{k=1}^{m} \alpha'_k \mathbf{1}_{\mathbb{R}_+}(-t - t_k), \qquad \alpha_k = r(t_k) - r(t_{k-1}),\ \ \alpha'_k = r(-t_k) - r(-t_{k-1}).$$

For $t \in [t_j, t_{j+1})$, $t \ge 0$, the second sum is 0 and the first telescopes to $r(t_j)$; symmetrically for $t < 0$. The grid point is within $1/m$, so

$$\sup_{t\in[-1,1]} |r(t) - s_m(t)| \le \frac{C_f}{m}.$$

Coefficients. Each $|\alpha_k|, |\alpha'_k| \le C_f/m$, so $S := \sum_k(|\alpha_k| + |\alpha'_k|) \le 2C_f \le 2C$.

Convex combination. $\mathbf{1}_{\mathbb{R}_+}(\langle \pm\mathbf{u}, \mathbf{x}\rangle - t_k)$ are half-space indicators; call the $2m$ coefficients β_j, indicators e_j. If $S = 0$, $s_m = 0 \in G_C$; otherwise

$$s_m(\langle \mathbf{u}, \cdot\rangle) = \sum_j \frac{|\beta_j|}{S}\Big(S\ \mathrm{sign}(\beta_j)\, e_j\Big),$$

weights $|\beta_j|/S$ summing to 1, brackets in G_C ($S \le 2C$): $s_m(\langle \mathbf{u}, \cdot\rangle) \in \mathrm{co}(G_C)$.

Limit. $\|g\|_{L^2_0(B_1^d)} \le \sup_{B_1^d} |g|$ (an average against a probability measure), so $\|k_{\boldsymbol{\xi}} - s_m(\langle \mathbf{u}, \cdot\rangle)\|_{L^2_0} \le C_f/m \to 0$, and $k_{\boldsymbol{\xi}} \in K$.

Step 3 (the average stays in K). The lecture's Riemann sum $\sum_{\boldsymbol{\theta}} k_{\boldsymbol{\theta}} \Lambda(I_{\boldsymbol{\theta}})$ over cubes $I_{\boldsymbol{\theta}}$ ($\boldsymbol{\theta} \in \frac{1}{n}\mathbb{Z}^d$) needs truncation and continuity of $\boldsymbol{\xi} \mapsto k_{\boldsymbol{\xi}}$ (the phase may jump where $\hat{f} = 0$); the following avoids both. [Supplement]

Projection. h has a nearest point p in the closed convex K:

$$\langle h - p,\, k - p \rangle \le 0 \qquad \text{for all } k \in K.$$

Averaging. By (5), $h - p = \int (k_{\boldsymbol{\xi}} - p)\, d\Lambda(\boldsymbol{\xi})$. Pair with $h - p$ and use Fubini ($|k_{\boldsymbol{\xi}}| \le C_f$ since $|r(t)| \le C_f |t|$; L^2 functions on the ball are integrable); by Step 2,

$$\|h - p\|_{L^2_0}^2 = \int_{\mathbb{R}^d} \langle h - p,\, k_{\boldsymbol{\xi}} - p \rangle \, d\Lambda(\boldsymbol{\xi}) \le 0.$$

Hence $h = p \in K$. ■

5.3 Completing the proof of Theorem 1

Proof of Barron's theorem. Lemma 8.16 and Lemma 8.15 ($H = L^2_0(B^d_1)$, $B = 2C$) give, for $c > 4C^2$, $|\gamma_i| \le 2C$, $\mathbf{a}_i$, b_i with

$$\frac{1}{|B^d_1|} \int_{B^d_1} \Big| f(\mathbf{x}) - f(\mathbf{0}) - \frac{1}{N} \sum_{i=1}^{N} \gamma_i \mathbf{1}_{\mathbb{R}_+}\big(\langle \mathbf{a}_i, \mathbf{x} \rangle + b_i\big) \Big|^2 d\mathbf{x} < \frac{c}{N}.$$

$\sigma(\lambda t) \to \mathbf{1}_{\mathbb{R}_+}(t)$ for $t \neq 0$, and the hyperplanes $\{\langle \mathbf{a}_i, \mathbf{x} \rangle + b_i = 0\}$ are null (for $\mathbf{a}_i = \mathbf{0}$, $b_i = 0$ the term is the constant γ_i, also given by $b_i = 1$). [Supplement]By dominated convergence, $\sigma\big(\lambda(\langle \mathbf{a}_i, \mathbf{x} \rangle + b_i)\big)$ with large λ keeps the error below c/N:

$$f_N(\mathbf{x}) = f(\mathbf{0}) + \sum_{i=1}^{N} \frac{\gamma_i}{N}\, \sigma\big(\lambda \langle \mathbf{a}_i, \mathbf{x} \rangle + \lambda b_i\big)$$

has N neurons and output bias $f(\mathbf{0})$. ■

6 Generalization and Rademacher Complexity

6.1 Population and empirical risk

With iid $\{(\mathbf{x}_i, y_i)\}_{i=1}^n \sim P$, loss ℓ and class $\mathcal{F}$ we want

$$\arg\min_{f \in \mathcal{F}} \; \mathbb{E}_{(\mathbf{x},y)\sim P}\big[\ell(y, f(\mathbf{x}))\big] \quad \text{(population risk)},$$

but can only compute

$$\arg\min_{f \in \mathcal{F}} \; \frac{1}{n} \sum_{i=1}^{n} \ell(y_i, f(\mathbf{x}_i)) \quad \text{(empirical risk)}.$$

Generalization error is controlled by $\sup_{f \in \mathcal{F}} \big| \frac{1}{n} \sum_i f(\mathbf{z}_i) - \mathbb{E} f \big|$ over the losses $\mathbf{z} = (\mathbf{x}, y) \mapsto \ell(y, f(\mathbf{x}))$.

6.2 Rademacher complexity

Definition 8.17: Rademacher variables and complexity

Random variables $\xi_1, \dots, \xi_n$ are **Rademacher** if they are iid with $P(\xi_i = 1) = P(\xi_i = -1) = \tfrac{1}{2}$. For a class $\mathcal{F}$ and iid samples $\mathbf{z}_1, \dots, \mathbf{z}_n$, the **Rademacher complexity** is

$$\mathrm{Rad}_n(\mathcal{F}) = \mathbb{E}\Big[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \xi_i f(\mathbf{z}_i)\Big],$$

the expectation being over both the ξ_i and the $\mathbf{z}_i$.

It measures how well the class fits random signs.

Theorem 8.18: uniform deviation bound

By symmetrization, $\mathbb{E}\sup_{f \in \mathcal{F}} \left|\frac{1}{n}\sum_i f(\mathbf{z}_i) - \mathbb{E}f\right| \le 2\,\mathrm{Rad}_n(\mathcal{F} \cup -\mathcal{F})$. If moreover every $f \in \mathcal{F}$ takes values in $[0,1]$, then with probability at least $1 - \delta$

$$\sup_{f \in \mathcal{F}} \Big|\frac{1}{n}\sum_{i=1}^{n} f(\mathbf{z}_i) - \mathbb{E}f\Big| \le 2\,\mathrm{Rad}_n(\mathcal{F} \cup -\mathcal{F}) + \sqrt{\frac{\log(2/\delta)}{2n}}.$$

(Lecture: "$\le 2\,\mathrm{Rad}(\mathcal{F}) + 1/\sqrt{2n}$", confidence factor suppressed.)

6.3 Two-layer networks

For $\mathcal{F} = \{f_{\boldsymbol{\theta}}(\mathbf{x}) = \sum_j \beta_j \sigma(\mathbf{w}_j^\top \mathbf{x})\}$ with $\sum_j |\beta_j| \le Q$, $\|\mathbf{w}_j\| \le W$, inputs $\|\mathbf{x}\| \le R$ and σ 1-Lipschitz with $\sigma(0) = 0$ (e.g. tanh), the contraction lemma gives

$$\mathrm{Rad}_n(\mathcal{F}) \le \frac{2QWR}{\sqrt{n}} = O\Big(\frac{1}{\sqrt{n}}\Big).$$

It depends on Q, not the number of neurons. With Barron: approximation $1/N$, generalization $1/\sqrt{n}$, both free of d.